Beyond the Foreground Screen

ULTIMATE–Subaru: machine-learning dust geometry

Question: what dust-related information survives realistic imaging?

Verified pilot: five views of TNG50 subhalo 294866

Bahareh | PhD progress / ULTIMATE workshop | November 2026

First establish a reproducible measurement

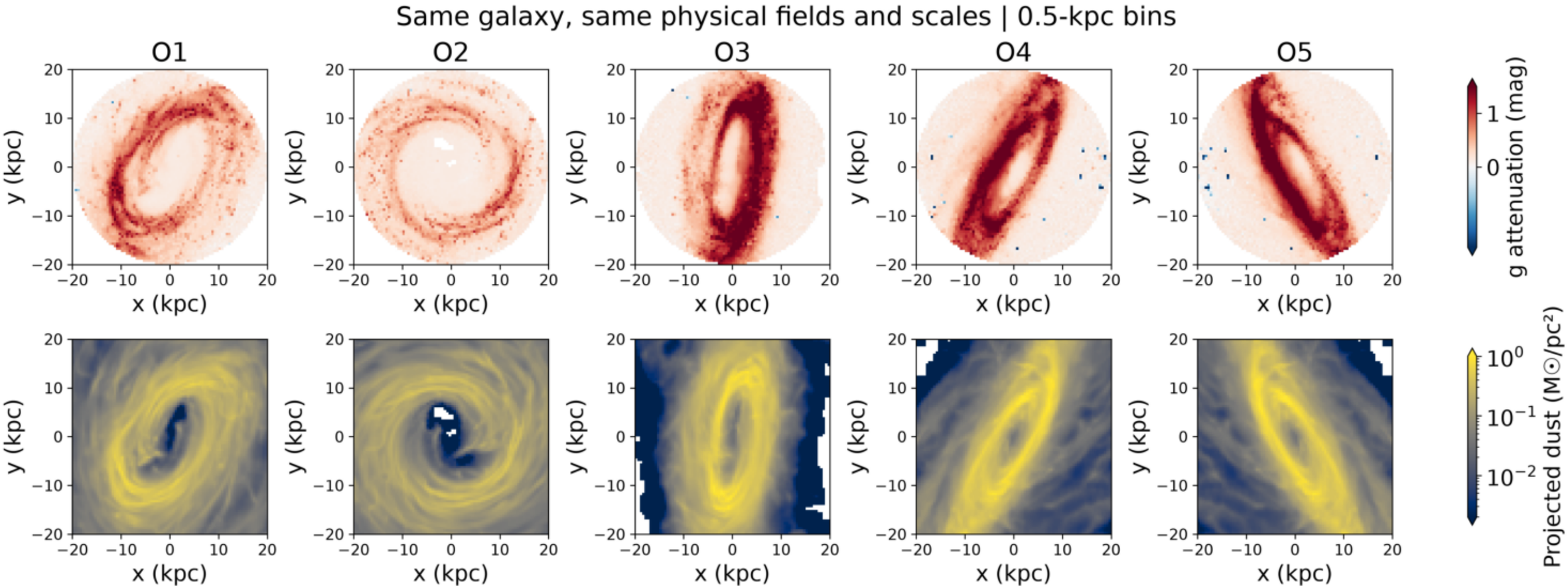
$A\lambda = -2.5 \log_{10}(I_{\text{with dust}} / I_{\text{without dust}})$

O1 reproduced: Ag = 0.334442 mag; AKs = 0.020562 mag

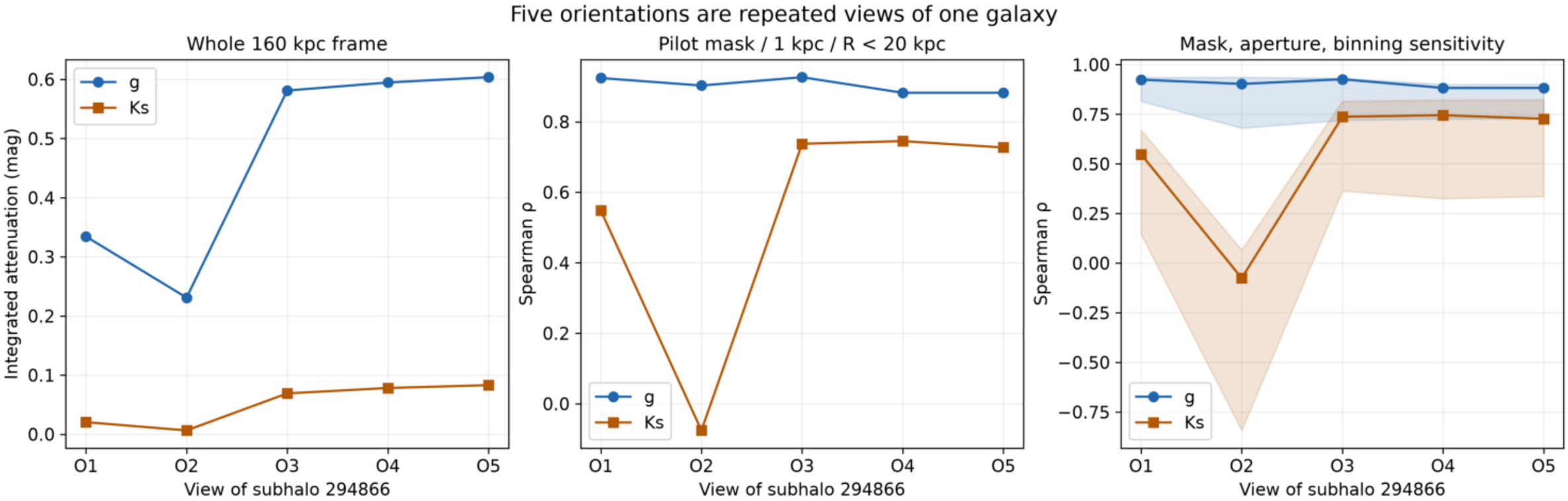
Spatial test: 1-kpc means, R < 20 kpc, shared faint mask

ρg = 0.924164; ρKs = 0.548340; 1,264 bins

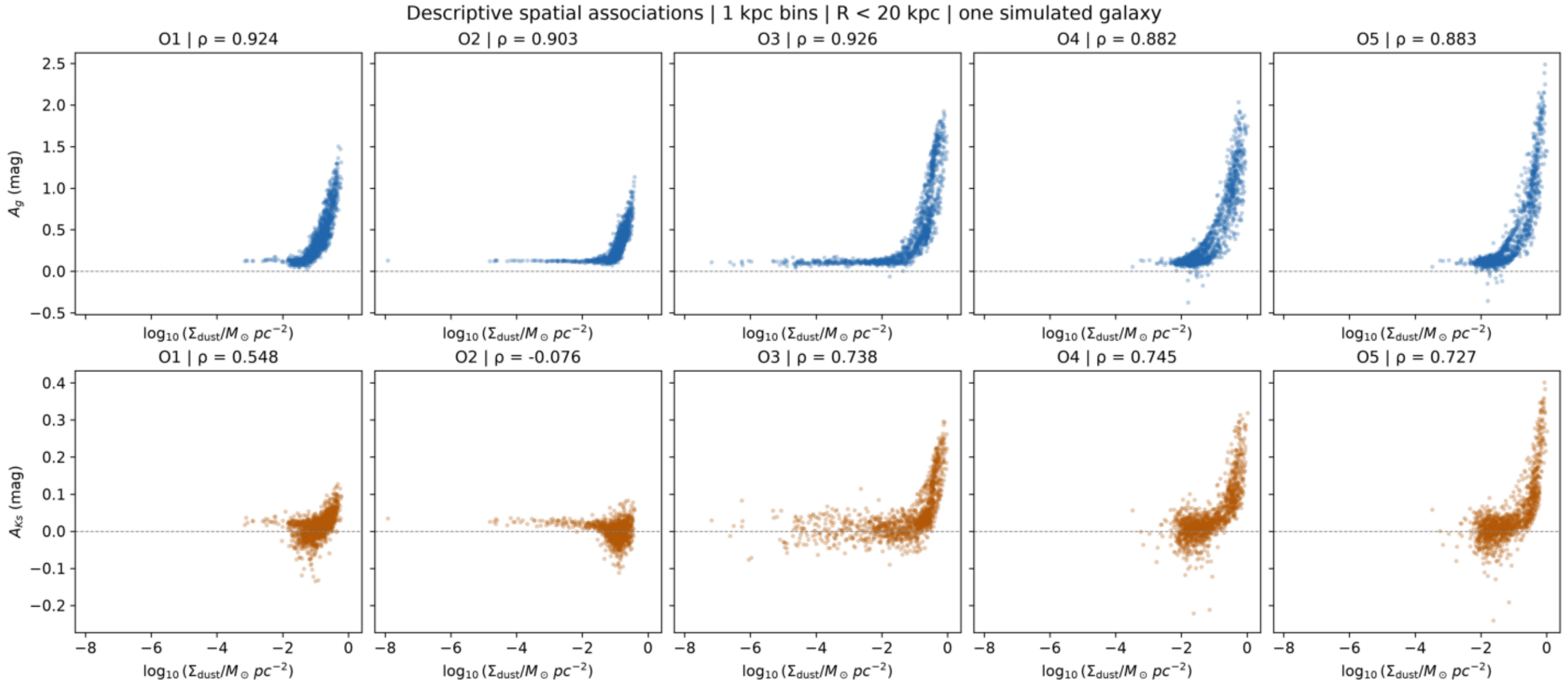
The projection changes across viewing directions



Integrated attenuation and association depend on view



Ks is not a universal monotonic dust tracer



Real-data anchor: raw HAWK-I/GRAAL Ks observations

22/22 exposures complete; raw-byte checksums and data sums pass

2018 January 3 and 6 | DIT × NDIT = 1.7 × 18 = 30.6 s

Four 2048² chips per frame | 0.10643 arcsec/pixel

Provisional frame median widths: 0.258–0.417 arcsec
Calibration, sky subtraction and final-stack PSF still required

Measured on raw frames | provisional PSF estimates | see HAWK-I inspection report

Next: a small forward-modelling feasibility test

Convolve intensities → angular scaling → pixel integration → noise

Design scenarios: K-band FWHM 0.2–0.3 arcsec; WFI 0.1 arcsec/pixel

Resolution limit: 100 pc at assumed DA = 100 Mpc is 0.206 arcsec
Finer output pixels cannot supply unresolved structure

Test observable colours against radius / brightness baselines
Filter curves and a verified noise budget are still needed

Planned, not executed | design references: ultimate.naoj.org/instrument.html; WFI-Motohara-v2s.pdf

What is established — and the next decision

Established: reproducible attenuation and viewing dependence

Established: complete HAWK-I data; provisional measured widths

Open: recoverable dust information at realistic depth and resolution

Then: independent galaxies and explicit 3D geometry labels
No large training run before this feasibility gate